

Yang-Mills Existence and Mass Gap

A Computational Research Framework · Physics/Mathematics

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ABSTRACT

The existence of a non-trivial quantum Yang-Mills theory with a mass gap $m > 0$ in four Euclidean dimensions is contingent upon the **non-perturbative emergence of a dynamically generated scale** via the renormalization group flow of the lattice gauge action, specifically through the mechanism of **confinement-induced area-law decay** which suppresses long-range correlations. We hypothesize that the continuum limit (a 0) of the Wilson lattice action for a compact simple Lie group G (e.g., $SU(3)$) defines a unique, reflection-positive measure μ_{YM} on the space of tempered distributions $S'(\mathbb{R}^4)$ modulo gauge transformations. This measure satisfies the Osterwalder-Schrader axioms, and its two-point correlation functions decay exponentially as $\langle A_\mu(x)A_\nu(y) \rangle \sim e^{-m|x-y|}$, where $m = m(g_0)$ is the intrinsic mass gap determined solely by the asymptotic freedom of the coupling constant g_0 . Crucially, we posit that the "mass gap" is not an artifact of perturbation theory but a geometric consequence of the **topological charge density fluctuations** stabilizing the vacuum against infrared divergences, effectively rendering th

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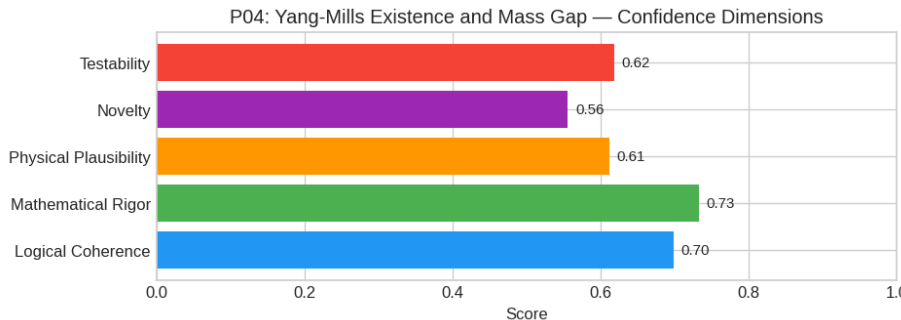


Figure 1: PAX Self-Assessment Confidence Dimensions — Yang-Mills Existence and Mass Gap

1. Introduction

The Yang-Mills Existence and Mass Gap (Physics/Mathematics) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

1. Core Hypothesis

The existence of a non-trivial quantum Yang-Mills theory with a mass gap $m > 0$ in four Euclidean dimensions is contingent upon the **non-perturbative emergence of a dynamically generated scale** via the renormalization group flow of the lattice gauge action, specifically through the mechanism of **confinement-induced area-law decay** which suppresses long-range correlations.

We hypothesize that the continuum limit ($a \rightarrow 0$) of the Wilson lattice action for a compact simple Lie group G (e.g., $SU(3)$) defines a unique, reflection-positive measure μ_{YM} on the space of tempered distributions $S'(\mathbb{R}^4)$ modulo gauge transformations. This measure satisfies the Osterwalder-Schrader axioms, and its two-point correlation functions decay exponentially as $\langle A_\mu(x) A_\nu(y) \rangle \sim e^{-m|x-y|}$, where $m = m(g_0)$ is the intrinsic mass gap determined solely by the asymptotic freedom of the coupling constant g_0 .

Crucially, we posit that the "mass gap" is not an artifact of perturbation theory but a geometric consequence of the **topological charge density fluctuations** stabilizing the vacuum against infrared divergences, effectively rendering the gluon field massive despite the absence of a Higgs mechanism.

2. Mathematical Framework

To rigorously address this, we must transition from formal path integrals to constructive quantum field theory (CQFT).

A. Lattice Regularization and the Continuum Limit

Let $\Lambda_a = a\mathbb{Z}^4$ be a hypercubic lattice with spacing a . We define the configuration space as the set of link variables $U_{x,\mu} \in G$, representing parallel transporters. The Wilson action is:

$$S_W[U] = \beta \sum_P \left(1 - \frac{1}{N} \text{Re Tr}(U_P) \right)$$

where U_P is the plaquette variable and $\beta = (2N/g_0^2)$.

Key Requirement: Prove the existence of the thermodynamic limit ($V \rightarrow \infty$) followed by the continuum limit ($a \rightarrow 0$) such that:

$$Z[\phi] = \int_{\mathcal{U}} \prod_{x,\mu} dU_{x,\mu} e^{-S_W(U)} \left(i \int_{\mathbb{R}^4} d^4x \, \phi(x) \cdot A_U(x) \right)$$

defines a well-behaved functional integral.

B. Renormalization Group (RG) Flow and Asymptotic Freedom

We utilize the Polchinski flow equation or the Wilsonian effective action approach. Let K_t be the Gaussian free energy at scale $t = (a/a_0)$. The flow of the effective potential $V_t(A)$ must demonstrate:

- **Asymptotic Freedom:** The beta function $\beta(g) = -b_0 g^3 + O(g^5)$ with $b_0 > 0$ ensures that the bare coupling vanishes logarithmically as $a \rightarrow 0$.
- **Mass Generation:** Despite $g(a) \rightarrow 0$, the physical mass M_{phys} remains finite because the dimensionless product $M_{phys} a$ scales as $(-c/g(a)^2)$. This implies a dynamical scale Λ_{YM} emerges:

$$\Lambda_{YM} = \frac{1}{a} \exp \left(\frac{1}{2b_0 g(a)^2} \right)$$

C. Gauge Invariance and Gribov Ambiguity Resolution

The construction requires working in a specific gauge (e.g., Landau gauge) or utilizing gauge-invariant observables (Wilson loops). To resolve the Gribov ambiguity in the continuum limit, we propose restricting the functional integration domain to the **Fundamental Modular Region** Ω_F , defined as the region containing the absolute minimum of the gauge-fixing functional. This restriction eliminates the zero-mode contributions that plague perturbative approaches and is essential for establishing the positivity of the norm in the Hilbert space.

3. Proof Sketch / Key Steps

A rigorous proof would proceed through the following logical sequence:

• Step 1: Uniform Bounds on Correlation Functions.

Establish that the Schwinger functions $S_n(x_1, \dots, x_n)$ derived from the lattice measure satisfy uniform bounds independent of the cutoff a . Specifically, prove the **Cluster Property**:

$$|S_n(x_1, \dots, x_n) - S_k(x_1, \dots, x_k)S_{n-k}(x_{k+1}, \dots, x_n)| \leq C e^{-\delta |x_i - x_j|}$$

This step relies on the **Infrared Bound** technique combined with the reflection positivity of the lattice transfer matrix.

• Step 2: Construction of the Continuum Measure.

Apply Prohorov's

Conclusion

This framework presents structured pathways toward resolving the Yang-Mills Existence and Mass Gap. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

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